01 Deriving Mole Balances for Common Reactors

Why are we learning this?

In the previous lesson, we introduced the **General Balance Equation** and learned that it is simply an accounting statement. Today, we will apply that accounting statement to some of the most common reactor types used in chemical engineering:

- Continuous Stirred Tank Reactors (CSTRs)
- Batch Reactors
- Plug Flow Reactors (PFRs)

Each reactor behaves differently because each reactor makes different physical assumptions.

By the end of today's lesson, you should understand:

1. How to derive a mole balance for a CSTR.
2. How to derive a mole balance for a Batch Reactor.
3. How to derive a mole balance for a PFR.
4. How reactor assumptions determine the form of the design equation.
5. Why all reactor design equations originate from the same conservation principle.

Course Roadmap

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Challenging the Well-Mixed Assumption
(Residence Time Distributions)
↓
Challenging the Elementary Rate Law Assumption
(Reaction Mechanisms)

Today's Location: Mole Balances for Common Reactors

Big Ideas

Every reactor design equation in this course begins with:

In − Out + Generation = Accumulation

The reactor equations may look different, but they all come from the same balance equation.

The differences arise because different reactors make different assumptions.

Reactor	Typical Design Question
CSTR	How large should the reactor be?
Batch Reactor	How long should the reaction run?
PFR	How large should the reactor be?

Today, we will see how those different questions emerge from the same conservation principle.

Let's start with a "warm up" example from Material and Energy Balances:

The reaction 2A + B --> 2C is being carried out in a reactor. A, B, and C are fed at rates of 100 mol/s, 200 mol/s, and 50 mol/s, respectively. G_A = -5 mol A/s. Determine F_C.

Accumulation = In − Out + Gen

0

assume Steady state

write the balance on any species

- we have $G_A = -5\frac{mol}{s}$ And we want FC.

So, pick A or C.

Balance on C:

$$0 = F_{C,in} - F_{C,out} + G_{net,C}$$

$$0 = 50\frac{mol\ C}{s} - F_{C,out}$$

$$+(-5\frac{mol A}{5})(\frac{2molC}{-2molA})$$

$$\therefore\ 0 = 50\frac{mol\ C}{s} - F_{C,out} + 5\frac{mol\ C}{s}$$

$$\boxed{F_{C,out} = 55\frac{mol\ c}{s}}$$

moles or molecules

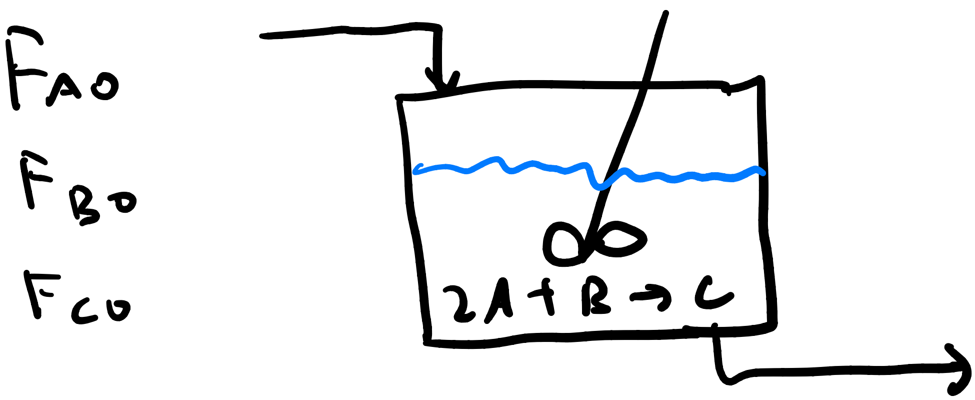
A Mole Balance Is Just Conservation of ~~Matter~~

General Balance: In – Out + Generation = Accumulation

This equation works for every reactor we study in this course. The differences between reactor design equations come from deciding which terms are zero and how we express the generation term.

Let's derive the mole balance for the CSTR.

- What is a CSTR and how does a CSTR work? [University of Michigan Equipment Encyclopedia: CSTR]



F_A

F_B

F_C

- <u>CSTR Key Assumptions:</u>
  - One: well mixed
  - Two: steady state
  - Three: operates at the exit conditions
- Taking these into account, the balance becomes:

$$0 = F_{j,in} - F_{j,out} + G_j\big|_{exit}$$

$$G_j\big|_{exit} = r_j\big|_{exit} V$$ = rate of generation of species j times the system volume

therefore $$0 = F_{j,in} - F_{j,out} + r_j\big|_{exit} V$$

- What variable do we solve for?

typically we solve for V $$V = \frac{F_{j0} - F_j}{-r_j\big|_{exit}}$$

What Makes a CSTR Different?

Physical Picture

A Continuous Stirred Tank Reactor is assumed to be perfectly mixed. As a result, conditions inside the reactor are the same as conditions at the outlet.

Engineering Consequence

When evaluating a rate law in a CSTR,

$$r_A = f(C_A, T, \ldots)$$

the concentrations and temperature are outlet values.

Remember

For a CSTR: One reactor volume → one set of conditions everywhere.

Example: The liquid phase reaction 2A --> B is being carried out isothermally in a CSTR. The rate law is kC_A^2 and the rate constant equals 0.03 L/mol*s. Species A is fed to the reactor at a concentration of 2 M and a flowrate of 3 L/s. Species A exits the reactor at a concentration of 0.1 M. Determine:

- The rate of reaction in the CSTR:

rate $= kC_A^2\big|_{exit}$

the rate is equal to the rate of consumption of species A which is equal to two times the rate of generation of species B

$-r_A =$ rate of consumption of species A $= kC_A^2\big|_{exit}$

$$-r_A = (0.03\frac{L}{mol \cdot s})(0.1\frac{mol}{L})^2 = 0.0003\frac{mol}{L \cdot s}$$

- The volume of the CSTR:

use the mole balance can write the mole balance on any species.

use species A since we already know $-r_A\big|_{exit}$

$$V = \frac{F_{AO} - F_A}{-r_A\big|_{exit}}$$ need FAO and FA

volumetric flowrate

unit conversion: $F_j = c_j v$

therefore, $F_{AO} = (2\frac{mol}{L})(3\frac{L}{S}) = 6\frac{mol}{S}$

$$F_A = (0.1\frac{mol}{L})(3\frac{L}{S}) = 0.3\frac{mol}{S}$$

then

$$V = \frac{6\frac{mol}{s} - 0.3\frac{mol}{s}}{0.0003\frac{mol}{L\cdot s}} = \text{19000 L}$$

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Now let's derive the balance for the batch reactor:

- What is a batch reactor and how does a batch reactor work? [University of Michigan Equipment Encyclopedia: Batch reactor]

1. fill the reactor
2. mix contents so that the system is well-mixed
3. carry out the reaction ← design equations
4. drain
5. clean

- Batch reactor key assumptions:
  - One: well mixed
  - Two: no inflows or outflows
- Taking these into account, the balance becomes: $Acc = \cancel{In} - \cancel{Out} + Gen$

Accumulation Of species A

$$\frac{dN_A}{dt} = r_A V$$

the accumulation of Species A = the rate Of generation of species A times system volume

- Special case: Constant volume batch reactor:

$$\frac{dN_A}{dt} = r_A V$$

V is NOT a function of time

$\div$ by V $\rightarrow$ $$\frac{1}{V}\frac{dN_A}{dt} = r_A$$

$$\frac{dC_A}{dt} = r_A$$

CA = concentration of A $= \dfrac{N_A}{V}$

function of T and CA

can only combine NA and V Since V is constant with t

What Makes a Batch Reactor Different?

Physical Picture

A batch reactor is filled, closed, and then allowed to react.

Engineering Consequence

There is:

- no inlet stream,
- no outlet stream,
- only reaction and accumulation.

Key Question

For a batch reactor we usually ask: How long must we wait? Time, rather than volume, becomes the primary design variable.

Let's do the same example as above, but this time for a batch reactor:

The liquid phase reaction 2A --> B is being carried out isothermally in a batch reactor. The rate law is kC_A^2 and the rate constant equals 0.03 L/mol*s. Species A is present initially in the reactor at a concentration of 2 M. Determine the time needed to decrease the concentration of species A to 0.1 M.

- Reaction: 2A --> B
- Reactor: Isothermal, constant volume batch reactor
- Rate law: kC_A^2, k = 0.03 L/mol*s.
- Initial concentration of A = 2 M.

- What variable do we solve for in the batch reactor?

time to achieve some conversion

- Determine: The batch time needed to *achieve* C_A = 0.1 M.

- The rate of reaction in the batch reactor is a function of time

- Applying and solving the batch reactor mole balance:

$$\frac{dN_A}{dt} = r_A V$$

or for constant V:

$$\frac{dC_A}{dt} = r_A$$

the rate of change in the concentration of A = the rate of generation of A.

$$\frac{dC_A}{dt} = r_A = -kC_A^2$$

$$\int_{2M}^{0.1M} \frac{dC_A}{C_A^2} = -k\int_0^t dt$$

$$\int_{C_{AO}=0.1M}^{C_A=0.1M} \frac{dC_A}{C_A^2} = \underbrace{-\frac{1}{0.1}+\frac{1}{2}}_{\text{units} = \frac{L}{mol}} = \underbrace{-kt}_{\text{units} = \frac{L}{mol \cdot s}}$$

$$\boxed{t = 317s}$$

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Constant Volume Batch Reactors

Why This Matters

A constant-volume batch reactor allows us to directly relate:

$$\frac{dN_A}{dt} = V\frac{dC_A}{dt}$$

because V does not change.

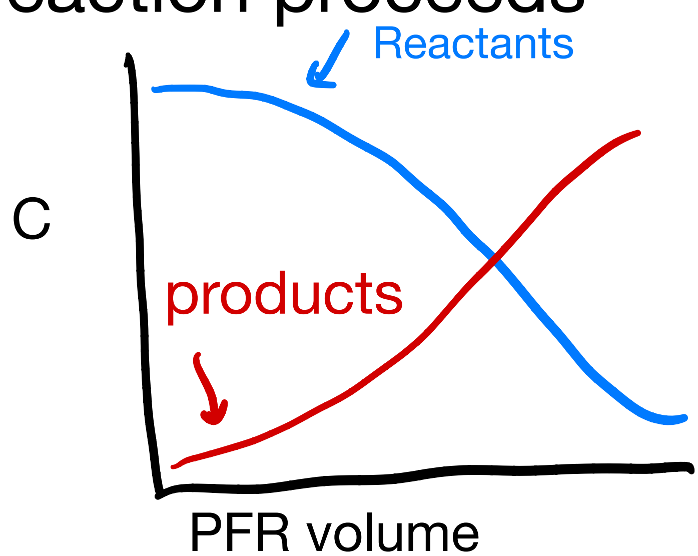
Result: The mole balance becomes easier to solve and leads to a differential equation involving concentration and time.

Check Yourself: If volume changes significantly, can you still pull V outside the derivative?

Let's derive the mole balance for the PFR.

- What is a PFR and how does a PFR work? [University of Michigan Equipment Encyclopedia: PFR]

a tube or pipe-fluid flows through it continuously -as the fluid progresses along the reactor, the reaction proceeds



- PFR Key Assumptions:
    - One:

Steady state

- Two:

perfect mixing in the radial dimension

Two a) turbulent flow aka plug flow.

- Three: continuous variation in the concentrations vs PFR length

- Like for the batch reactor,
  concentration varies
  but different than the batch reactor
  it varies w/ position NOT time

- How do we define the system?

  - ~~If we know G_A:~~

  - ~~If we do not know G_A:~~

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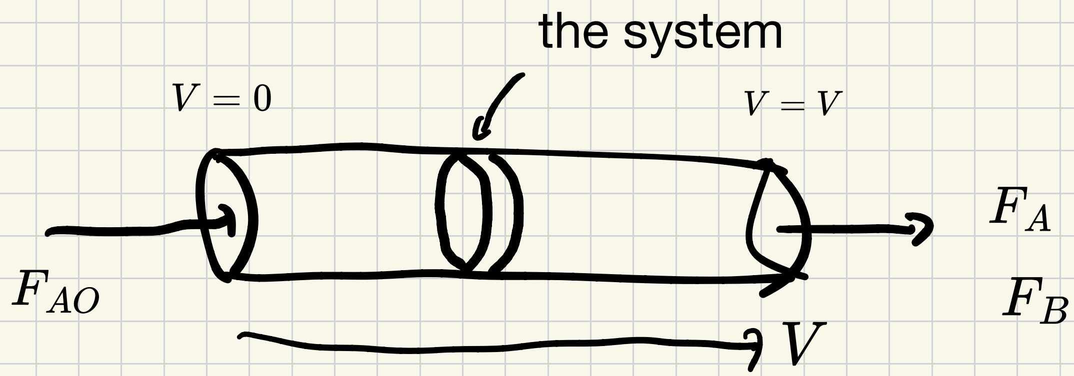
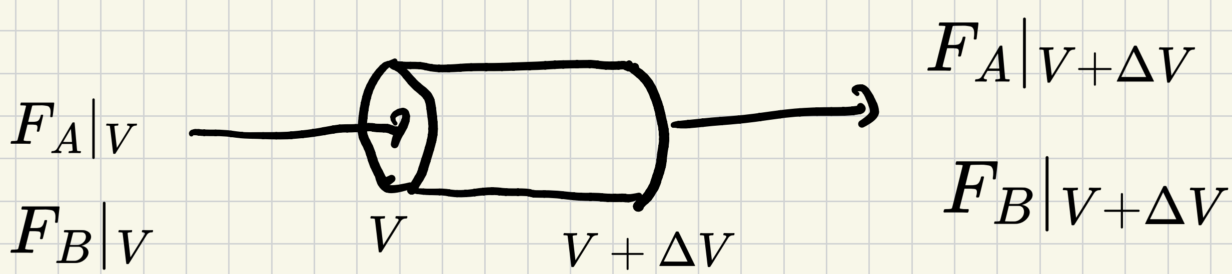


1. define our system -define as an infinitesimally small volume slice
-do this b/c concentration in that slice is uniform by definition

Zoom in on system:



- write a balance on species A

Accumulation = In - Out + Gen

$$0 = F_A|_V - F_A|_{V+\Delta V} + r_A \Delta V$$

volume of the system

$\div \Delta V$ & take the limit
as $\Delta V \to 0$

Apply the definition of the derivative. Divide by ΔV and take the limit as $\Delta V \to 0$. (This works perfectly with our assumption that ΔV is infinitessimally small.) Then:

$$\underbrace{lim_{\Delta V \to 0} \frac{F_A|_V - F_A|_{V+\Delta V}}{\Delta V}}_{-1\times \text{ definition of a derivative}} = -r_A$$

$$-\frac{dF_A}{dV} = -r_A \rightsquigarrow \boxed{\frac{dF_A}{dV} = r_A}$$

in words: the change in the molar flowrate of A with respect to volume = rate of generation of A

What Makes a PFR Different?

Physical Picture

Fluid moves continuously through a tube.

Material entering the reactor encounters different conditions as it travels downstream.

Engineering Consequence

Concentration, temperature, and reaction rate vary with position.

Remember

For a PFR:

One reactor volume → many different conditions.

This is different than a CSTR.

Why Use a Differential Volume in a PFR?

Big Idea

A PFR does not have one uniform concentration.

Instead, conditions change continuously along the reactor length.

Solution

We analyze a very small slice of reactor volume, ΔV, and apply a mole balance to that slice.

Taking the limit as $\Delta V \rightarrow 0$ produces the differential form of the PFR design equation.

We then integrate over the ΔV to determine the total volume necessary.

Example: The liquid phase reaction 2A --> B is being carried out isothermally in a PFR. The rate law is kC_A^2 and the rate constant equals 0.03 L/mol*s. Species A is fed to the reactor at a concentration of 2 M and a flowrate of 3 L/s. Species A exits the reactor at a concentration of 0.1 M. Determine the volume of the PFR. Is this volume larger or smaller than the volume for the CSTR? Explain the difference.

$$\frac{dF_A}{dV} = r_A \quad , \quad r_A = -kC_A^2$$

$$\frac{dF_A}{dV} = -kC_A^2$$

$$F_A = C_A v \longrightarrow \left(\frac{mol}{L}\right)\left(\frac{L}{s}\right) = \frac{mol}{s}$$

v is constant

$$v\frac{dC_A}{dV} = -kC_A^2$$

$$\frac{dC_A}{C_A^2} = -\frac{k}{v}dV \longrightarrow \int_{2M}^{0.1M} \frac{dC_A}{C_A^2} = -\frac{kV}{v}$$

$$-\frac{1}{0.1} + \frac{1}{2} = -\frac{kV}{v} \qquad \boxed{V = 950L}$$

Summary

$$\frac{L}{mol} \qquad mol^{-1}$$

Today we derived:

Reactor	Design Equation Obtained By
CSTR	Applying the general mole balance to a single, well-mixed volume
Batch	Eliminating flow terms from the general mole balance and accounting for non-steady state behavior
PFR	Applying the general mole balance to a differential volume element

Key Takeaway: All reactor mole balances come from the same equation:

$$\text{In} - \text{Out} + \text{Generation} = \text{Accumulation}$$

The different reactor equations arise because different reactor types make different physical assumptions.

Key Takeaway: The chemistry determines the reaction rate. The reactor determines how that rate is used in the design calculation.

Question: CSTR volume

Is $19000L$ and PFR volume

is $950L$.

Why is the

PFR volume so much smaller?!